

Fuzzy Comfort-Aware RRT* Path Planning for Dynamic Human-Interactive Environments

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Abstract— Robotic navigation in human environments requires not only safety and efficiency but also comfort and social awareness. Traditional sampling-based algorithms, such as Rapidly Exploring Random Tree Star (RRT*), are effective for static obstacle avoidance but often struggle to adapt to dynamic, human-populated spaces while maintaining user comfort. This paper presents a hybrid comfort-aware navigation framework that integrates RRT* with a fuzzy logic-based comfort evaluation system. Unlike conventional RRT*-based planners that primarily optimize path feasibility and efficiency, the proposed framework incorporates real-time comfort assessment and adaptive local replanning to improve navigation behaviour in human-interactive environments. The method generates an initial global path using RRT*, while a fuzzy inference system continuously evaluates local comfort levels based on the robot's distance and relative velocity with respect to nearby moving entities. When the comfort score falls below a defined threshold, a lightweight roadmap-based local replanning module is triggered to restore a comfortable and safe trajectory. The approach was evaluated in a simulated classroom environment containing static furniture and dynamically moving obstacles representing human teachers. Experimental results demonstrate a 57% improvement in comfort score, an increase in average obstacle clearance from 1.84 m to 2.55 m, and a 6.6% reduction in path length compared with conventional RRT*, while maintaining an average replanning time of approximately 0.14 s. These results highlight the potential of the proposed framework to achieve socially aware, comfortable, and efficient robot navigation in dynamic indoor environments.

Keywords— RRT*, fuzzy logic, path planning, comfort-aware navigation, dynamic obstacles, human-robot interaction

I. INTRODUCTION

The gradual adoption of mobile robots in places like classrooms, hospitals, and offices where humans are the focus has resulted in the need for more advanced navigation systems that are capable of more than just simple obstacle avoidance and route efficiency. In such areas where people and robots coexist, robots are expected not only to be safe but also to move in a manner that is natural and comfortable for the humans present. Although traditional motion planning algorithms, including different forms of the Rapidly Exploring Random Tree Star (RRT*), have demonstrated huge potential for finding paths that do not cause collision in stationary environments, these methods generally consider obstacles as fixed and do not take into account the movements of dynamic agents like humans who might influence the overall comfort and safety through their motion patterns and proximity. That is why very often the generated paths will be perfectly fine from a geometric point of view; however, they might be invading personal spaces, or the motion might be discerned as being abrupt rather than smooth. To overcome this issue, the implementation of human-aware or comfort-aware decision

layers into robotic planning frameworks is becoming more and more popular. Fuzzy logic, for its part, can model both uncertainty and subjective perceptions, allowing the successful human comfort quantification via the linguistic rules framing human reasoning. A robot that relies on the structured exploration of RRT* and a fuzzy inference system that is able to assess environmental comfort in real time, can then modify its trajectory continuously as per the static and mobile entities. Henceforth, this paper discusses a hybrid fuzzy-RRT* method that aims at making path comfort and adaptability in dynamic indoor environments better. The intended system takes advantage of RRT* for the global path generation and a fuzzy comfort model that assesses spatial comfort based on the distance and relative velocity to the closest agents, thus allowing easy rerouting whenever comfort drops.

II. PROBLEM STATEMENT

Robots that can work independently are expected to move around conveniently and naturally in classrooms and hospitals, which are human shared environments. The evident advantages of traditional path planning methods like RRT* are that they can find collision-free routes, but are unable to adapt to subtle aspects of human comfort, like personal space and avoiding sudden or awkward manoeuvres around moving individuals. This leads to socially uncomfortable paths, although they are technically valid. Also, current rerouting strategies usually only focus on obstacle avoidance and do not consider how dynamic human movement affects safety perception and interaction ease. That's why the challenge is to create a planning approach that can mix the accuracy of geometric navigation with an adaptive comfort model, giving the robot the ability to constantly change its path depending on both still structures and human motion that cannot be predicted.

III. LITERATURE REVIEW

The early studies that concentrated on the use of sampling methods for path planning were the starting point for the efficient and rich generation of motion in complicated environments. Noreen et. al [1], [2] made an in-depth analysis of the classical RRT and of the different RRT* and RRT*-Smart variants, indicating an increase in convergence rate, smoother paths, and less computation. Their comparative and survey studies established RRT* as a versatile base framework for autonomous navigation in static and semi-dynamic situations concerning convergence, smoothness, and computational efficiency. After these principles, Connell and La [6] in their work took off RRT* for dynamic replanning, and were able to show that RRT* is capable of changing its path when in the presence of obstacles while still being optimal. Ganesan et al. [9] introduced G-RRT*, a goal-oriented modification of RRT* that pulls the sampling closer

to the end point and it is thus better at both convergence and reliability for mobile robots moving around in indoor spaces that are not very cluttered.

At the same time, other researchers conducted separate but related studies in which they incorporated smart control methods into the classic path-planning algorithms. The research groups of Davis and Supriya [4], Sandeep and Supriya [12] were the first to apply fuzzy logic to robotic path planning and they demonstrated that the sensor data could be transformed into smooth adaptive motion decisions through the application of linguistic rules. Alomari et al. [5] used models of dynamic fuzzy logic for the solution of mobile localization problems and thus they were able to assure a robust navigation for the robot in environments that are both uncertain and subject to sensor-noise. A similar extension of the concept was made by Kumar et al. [11] who suggested a multi-agent planning approach based on fuzzy support vector machine and dealing with cooperating robot obstacle avoidance and optimizing interaction areas. Song et al. [3] reported the result of merging fuzzy reasoning with ant colony optimization, giving backing to the idea that hybrid fuzzy systems are the ones that make real-time adaptation possible in the challenging dynamic environments where precise models are hard to come by.

In the very near future, the researchers put a bigger emphasis on a socially and comfort-aware navigation of robots in human-inhabited places. To this end, Truong et al. [10] proposed a socially aware robot navigation system that can tell the areas of human interaction and Singamaneni et al. [8] offered a thorough taxonomy of social navigation techniques with an emphasis on personal-space modelling and behavioural prediction of a dynamic nature. Wang et al. [7] resorted to globally-guided reinforcement learning in order to enhance the robot's decision-making capability in the presence of moving obstacles and thus managed to understand the context of these obstacles. All the mentioned works can be regarded as indicative of the shift from purely geometric and optimization-based planning towards human-centered, comfort-aware navigation frameworks. The literature calls for a hybrid architecture combining the RRT* geometric precision with the fuzzy logic adaptive reasoning—so as to seamlessly link efficient motion planning with real-time comfort-oriented rerouting in shared spaces.

Although some real progress has been made in robotic navigation thanks to sampling based planning, fuzzy control systems, and socially aware navigation frameworks, a few limitations still stay around. In many older RRT* style approaches, the main attention goes to path optimality and simple collision avoidance, but they often skip human comfort, and also whether the motion feels socially acceptable while moving around. Social force based methods do help with human aware behaviour, yet they usually need careful parameter tuning, and in very crowded places they can get computationally heavy. Reinforcement learning approaches bring adaptable decision making, though they often rely on huge training datasets and also on serious compute budgets to work properly. So in the end, there is still a need for a navigation framework that is computationally light, but still keeps path efficiency, obstacle avoidance, and human comfort all at once in dynamic environments. The proposed fuzzy comfort aware RRT* framework tries to fill this space by blending real time comfort assessment with adaptive local

replanning, while at the same time keeping the efficiency that sampling based path planning is known for.

IV. GLOBAL PATH PLANNING USING RRT*

Planning the global path is the first task to be done in the proposed navigation framework. Here, the task is to find a non-collision and optimal route in the 2D working area filled with static obstacles, between a given starting point and a goal point. The method used for this is Rapidly-Exploring Random Tree Star (RRT*), which is a widely accepted sampling-based planner that has the property of asymptotic optimality and has high dimensional spaces as its main advantage.

In the case of the RRT* algorithm, at every iteration the tree of feasible configurations $T=\{x_i\}$ is incrementally composed of a random drawing of the configuration space for each of the existing nodes and directing the sampling towards unexplored areas. A random point x_{rand} is chosen within the limits of the workspace for every iteration. The nearest existing node x_{near} is identified by minimizing the Euclidean distance:

$$x_{near} = \arg \min_{x_i \in T} \|x_{rand} - x_i\|$$

A new point x_{new} is then generated by moving from x_{near} toward x_{rand} with a fixed step size η :

$$x_{new} = x_{near} + \eta \cdot \frac{(x_{rand} - x_{near})}{\|x_{rand} - x_{near}\|}$$

The connection between x_{near} and x_{new} is only accepted if the line segment does not intersect any obstacle region. This ensures that every edge in the resulting tree corresponds to a feasible, collision-free motion.

Once x_{new} is added to the tree, a local optimization step—called *rewiring*—is applied to improve the cumulative cost of neighbouring nodes. The total path cost from the start node x_{start} to a node x_i is given by:

$$J(x_i) = J(x_{parent}) + \|x_i - x_{parent}\|$$

where $J(x_i)$ denotes the accumulated path cost. By minimizing this cost iteratively across sampled nodes, the RRT* gradually converges toward the globally optimal solution.

After constructing the tree, the path connecting x_{start} to the goal node x_{goal} is extracted and post-processed using cubic B-spline interpolation to improve curvature continuity and smoothness:

$$P_{smooth}(t) = \sum_{k=0}^n B_k^3(t) \cdot x_k$$

where $B_k^3(t)$ represents the cubic B-spline basis functions and x_k are the sampled waypoints along the RRT* trajectory.

The path devised by RRT* globally shows the robot's baseline reference trajectory. It is the shortest and most geometrically feasible route but does not consider the higher-level factors such as human comfort, motion predictability, or dynamic agents being present. The above-mentioned factors are, however, later incorporated into the comfort-aware and local rerouting stages.

V. FUZZY-BASED COMFORT-AWARE REROUTING

Global path planning, while providing the best geometric path, does not take into consideration the mobile robot's comfort and safety margin in human-like environments. In a dynamic classroom consisting of teachers or moving obstacles, sticking to the original RRT* path can result in socially uncomfortable situations, such as the robot passing too close to people or entering cramped areas once again. To overcome this drawback, a fuzzy logic-based comfort evaluation system, which can influence real-time rerouting decisions, was incorporated into the navigation framework.

A. Fuzzy Comfort Evaluation

The comfort model quantifies the perceived comfort of the robot's motion based on its distance from nearby obstacles and their relative velocities. Two fuzzy input variables were defined: distance (d) and relative velocity (v_r), and one output variable, comfort (C).

The fuzzy membership functions were constructed as shown in Fig. 1:

$$d = \{\text{close, medium, far}\}$$

$$v_r = \{\text{approaching, still, leaving}\}$$

$$C = \{\text{unsafe, neutral, comfortable}\}$$

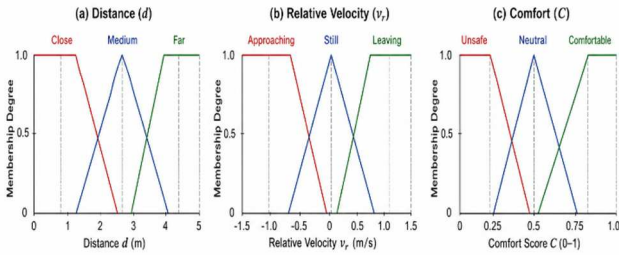


Fig. 1. Membership Functions for Fuzzy Comfort Evaluation

Smooth transitions between linguistic states were achieved using triangular and trapezoidal membership functions.

TABLE I. FUZZY RULE BASE FOR COMFORT EVALUATION

Distance	Relative Velocity		
	Approaching	Still	Leaving
Close	Unsafe	Unsafe	Neutral
Medium	Unsafe	Neutral	Comfortable
Far	Neutral	Comfortable	Comfortable

The fuzzy inference system evaluates comfort using a rule-based framework that combines distance and relative velocity information. The rule base was designed to reflect intuitive human comfort behaviour, where comfort decreases when a robot moves quickly in close proximity to a person and increases as the separation distance grows. Table 1 summarizes the fuzzy rules used in the proposed system.

The comfort output $C(x,y) \in [0,1]$ is obtained through a centroid defuzzification process, giving a continuous measure of comfort across the workspace. Regions near moving agents or obstacles yield lower comfort scores.

Mathematically, the comfort field is modelled as:

$$C(x, y) = f(d_{static}(x, y), d_{dynamic}(x, y), v_r)$$

where d_{static} is the distance to the nearest static obstacle, $d_{dynamic}$ is the distance to the nearest moving obstacle, and v_r represents the relative motion of the obstacle with respect to the robot.

B. Comfort-Aware Decisions and Rerouting

During navigation, the robot continuously evaluates its comfort score along the current path with the following formula:

$$C = \alpha \cdot \text{clip}(d_{min}/2, 0, 1) + (1 - \alpha)\eta$$

Where d_{min} is the minimum distance from the robot to the nearest obstacle (either static or dynamic) and η is the damping factor in the range $[0.9, 1.0]$, introducing stochastic variation to simulate real-world uncertainty and sensory noise. The weighting parameter α and the comfort threshold values were selected through preliminary simulation trials conducted under representative classroom navigation scenarios. Multiple parameter combinations were evaluated to balance navigation efficiency and human comfort. Values within the range of 0.6–0.8 produced stable performance, while $\alpha = 0.7$ provided the most consistent trade-off between path optimality and comfort preservation. Similarly, the selected comfort threshold enabled timely replanning without introducing excessive path deviations or computational overhead.

When the comfort value at the current position C_t drops below a defined threshold $C_{min} = 0.4$, or when a dynamic obstacle moves within a safety radius $r_{th} = 0.8$ m, a reroute trigger is initiated.

Rather than canceling the global RRT*, a fast local replanning scheme is employed to provide quick response. A Lazy Graph Planner builds a graph of safe points throughout the area in advance and connects them with edges that are checked for being free from both collision and comfort constraints. When a rerouting event occurs, the planner finds the least-cost path between the robot's present position and the target using a comfort-weighted Dijkstra method:

$$J_{replan}(p_i, p_j) = \|p_i - p_j\| * (1.2 - 0.5C(p_j))$$

In this scenario, the cost of each edge is inversely related to the level of comfort which implies that the robot will prefer the more comfortable regions with higher C , even if they are slightly longer. The dynamic cost scaling makes the robot reroute through the socially comfortable open spaces rather than being trapped between desks or dangerously close to people.

The rerouted path is further refined by a cubic spline to ensure the kinematics are still good, and this new part is represented as a blue dashed line in the classroom simulation. The system shows all paths that have been previously traveled or rerouted so as to demonstrate the planner's adaptive character.

C. Integration within the Navigation Loop

The fuzzy comfort assessment and rerouting logic are functioning all the time in the main navigation loop. The positions of all dynamic obstacles are updated at each time step, and the robot's comfort score C_t is computed with the help of the fuzzy inference system. The comfort level going

down a preset limit or a moving obstacle entering the safety radius causes the triggering of the rerouting event. The Lazy Graph Planner then creates a new local path based on the comfort-weighted cost function, thereby ensuring the quick adaptation without the need for recomputing the whole RRT* tree. The redirected part gets the cubic spline applied to it, is smoothed, and then added to the activated trajectory. This instantaneous operation allows the robot to perform a smooth, safe, and comfort-aware motion even when it has to deal with dynamically changing obstacles.

VI. RESULTS AND DISCUSSION

The method that was put forward was assessed in two phases: the first phase consisted of a benchmark test on the RRT* algorithm using a randomly generated obstacle field, and the second phase consisted of deploying it in a classroom environment that was organised in a way to simulate human-occupied indoor navigation. During the initial stage, the RRT* algorithm was able to show its capability of producing collision-free global paths for different obstacle configurations, as demonstrated in Fig. 2. This experiment ensured that the baseline planner was, for certain, able to navigate through the complicated areas and at the same time get to the optimal routes before the comfort-based rerouting is applied. The traces obtained via this testing confirm the claim that RRT* is not only smooth but also efficient in producing feasible paths. But, these baseline paths does not offer any consideration for comfort, proximity, or human-like navigation preferences as they were purely geometric.

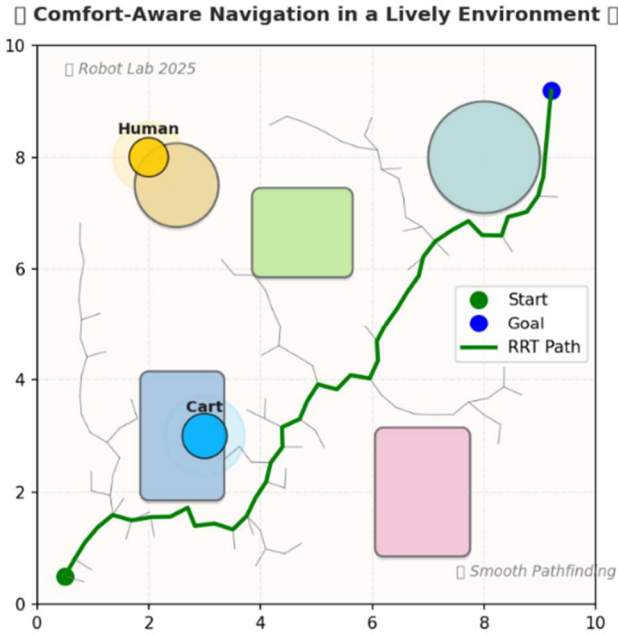


Fig. 2. Initial RRT* planning

To further pursue this project through social navigation, a classroom setting was created as shown in Fig. 3. The scenario had desks and chairs arranged in rows like static obstacles and two teachers represented by dynamic agents moving from one side to the other across the aisles. This arrangement was chosen because it is similar to the real-life situation of navigating through tight places where the movement and the distance between people are constantly changing.

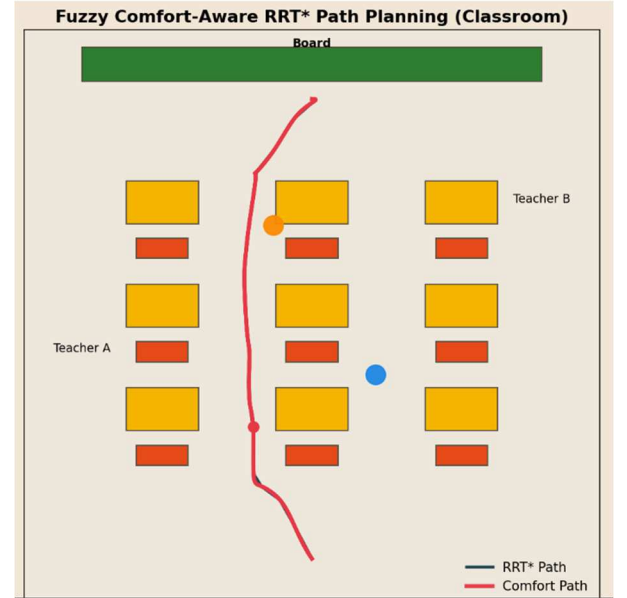


Fig. 3. Classroom Design Setup

Within this environment, the RRT*-generated global path served as the robot's initial route. As the teachers moved, their proximity triggered local drops in comfort level, computed through the fuzzy logic system. Once the comfort score fell below a set threshold, the robot initiated a fuzzy comfort-aware rerouting process, generating an alternative path through a lightweight graph-based planner. Fig. 4 shows that the reroutes, shown as blue dashed lines, were executed in real time, preserving both trajectory smoothness and workspace feasibility.

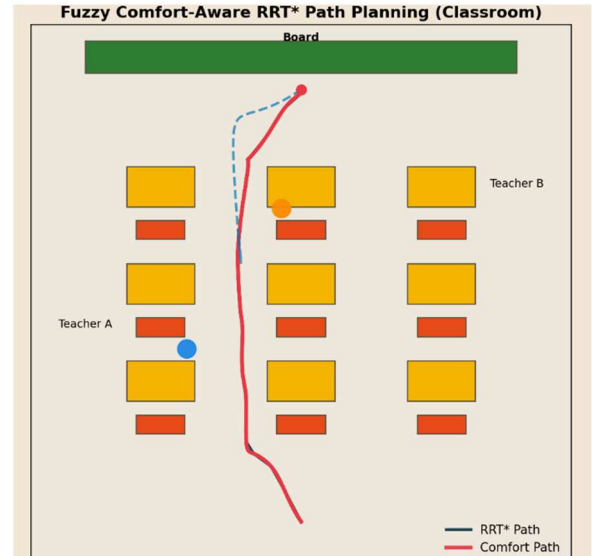


Fig. 4. Path rerouting shown with blue dashed line

The qualitative results further support this proposed approach. As illustrated in Fig. 5, the generated trajectories show smoother avoidance behaviour when the system faces dynamic obstacles. Instead of merely picking the shortest feasible route, the planner flexibly adapts its path, so it keeps a steadier buffer from nearby people, which then boosts the overall comfort levels. The comfort heatmaps in Fig. 6 also relay the same output, the robot keeps steering away from areas tied to higher discomfort, and so the navigation

behaviour ends up being more appropriate for human interactive environments.

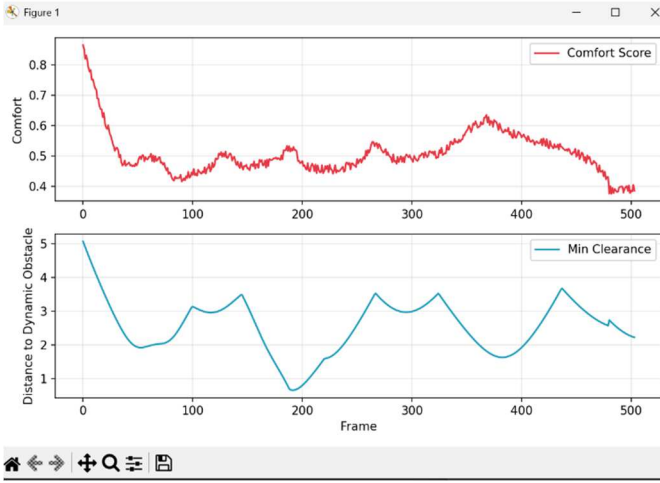


Fig. 5. Per-frame metrics

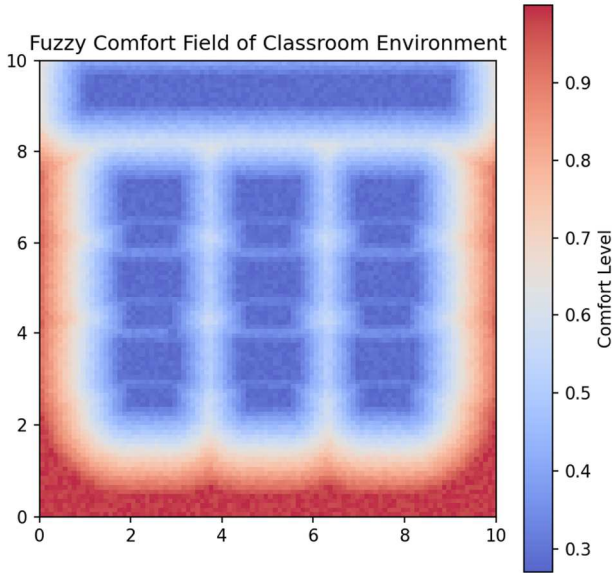


Fig. 6. Comfort Heatmapping

The quantitative results show that the proposed fuzzy comfort-aware framework does, overall, deliver a balanced gain across navigation efficiency and human-centric performance measures. If we compare it with the baseline RRT* planner shown in Table 2, the proposed approach gives a comfort score that jumps by 57% , boosts the average obstacle clearance from 1.84 m to 2.55 m, and at the same time cuts the total path length by 6.6%. So, all together, the planner really manages to favor human comfort, without taking away the navigation efficiency part.

Also the average replanning time is around 0.14 s, which points to it being suitable for real-time use in changing scenarios. The findings indicate that when the robot uses comfort-aware decision making, it can keep socially acceptable distances from nearby pedestrians, while still holding smooth and efficient motion paths.

TABLE II. PERFORMANCE COMPARISON BETWEEN BASELINE RRT* AND FUZZY COMFORT-AWARE RRT*

Metric	Baseline RRT*	Fuzzy-RRT*	Observation
Path Length	9.636 units	9.002 units	6.6 % shorter route
Average Smoothness	0.998	0.998	Maintained trajectory continuity
Average Comfort Score	0.322	0.508	57% increase in comfort metric
Average Clearance	1.84 m	2.55 m	Safer distance from obstacles maintained
Replans Triggered	-	1	Minimal reroutes indicate stable performance
Average Replan Time	-	0.139 s	Rapid, low-latency local replanning
Path Deviation	-	0.45 units	Limited divergence from the original path

In comparison with conventional RRT*-based navigation, the proposed framework brings an extra layer of comfort-aware reasoning via fuzzy logic, while not adding much computational burden. While learning-based approaches, like reinforcement learning, can deliver adaptive navigation behaviour, they usually hinge on lots of training data and heavy computation. By contrast, this method offers a more interpretable and computationally light solution, it can also be fused into existing navigation systems without too much hassle, while still enhancing human comfort and obstacle clearance.

The results, when combined, show that the application of fuzzy-based comfort reasoning has a significant impact on the navigation behavior in socially interactive environments, making the method endurable. The system not only avoids collisions but also adds human-like soothing, spatial comfort, and adaptability, thus, a perfect fit for real-world applications in places like classrooms, hospital corridors, and indoor service robotics.

VII. CONCLUSION AND FUTURE WORK

This paper presents a fuzzy comfort aware RRT* path planning framework for dynamic, human interactive settings. It blends the global optimality of RRT* with real time fuzzy comfort evaluation and adaptive local replanning, so as a whole the method allows safer navigation and more socially acceptable robot movement. In the experiments, the reported outcomes show higher comfort score, better obstacle clearance, and improved path efficiency, while still keeping replanning times low enough for real time use.

Even though the current study leans on simulation based validation, the obtained results still point to the promise of comfort-aware navigation strategies for service robots that work around people. Looking ahead, future work will concentrate on real world deployment, bringing in human behavior prediction models, and running evaluation in more complex, multi agent scenarios.

RRT* serves as the main technique to create an initial globally optimal path that not only provides coverage of the entire workspace but also ensures free connectivity between the robot and the obstacles. Then the pathway is fine-tuned by using a cubic spline interpolator so that there is a continuity in motion and a smooth curvature that is appropriate for the real-world mobile platforms. The fuzzy comfort model constantly

measures the distance and the relative motion between the robot and the other agents in the vicinity to derive a real-time comfort rating. If the rating goes below a certain limit or if fast-moving obstacles come into the safety zone, a rerouting process is initiated through the pre-established Lazy Graph Planner. The local replanner then fine-tunes the new path by solving an optimization problem with a comfort-weighted cost function, ensuring that the rerouting choices are made taking into account both the safety and the natural motion of the robot.

A comparison between the standard RRT* and the suggested fuzzy comfort-aware RRT* shows evident navigation performance enhancements. The objective measures suggest that the total path length has decreased, the average comfort level has increased, and the interactions with the dynamic entities have been smoother. The real-time rerouting traces and comfort variation maps, among the other visualizations, corroborate that the robot can lead a stable and socially distanced lifestyle and to cope with dynamically moving obstacles at the same time. The performance of the framework is quite solid when the classroom is simulated but still it is very much possible to unlock its latent potential. In the coming days or months, the focus of work will be on integrating the predictive motion models of dynamic agents and extending the comfort evaluation to multi-agent scenarios, whereby the human intention and crowd behaviors can be predicted. Adding learning-based comfort adaptation, actual deployment using mobile robots, and integration with vision-based perception modules will all contribute greatly to the system's autonomy and applicability. In short, the research effort provides a complete adaptive path planning model that combines classical RRT* exploration with fuzzy-based comfort reasoning. The system, by combining deterministic sampling, soft-decision logic, and real-time rerouting, achieves technical efficiency and socially aware motion planning, thus laying the groundwork for intelligent, human-centred navigation in dynamically changing indoor environments.

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